



Seismic Shielding



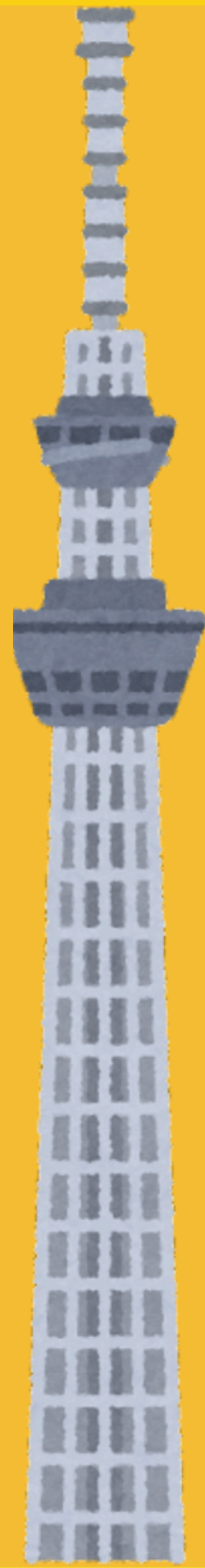
Tokyo Skytree

- 3rd Tallest Free Standing Structure in the world
- Located in one of the most seismically active regions on earth
- Has lasted for 14 years (Opened in 2012)

How has the Tokyo Skytree lasted so long?

It has a Shinbashira, a cylindrical column made of reinforced concrete, which is installed in a void in the core of Tokyo Skytree, structurally separate from the rest of the tower.

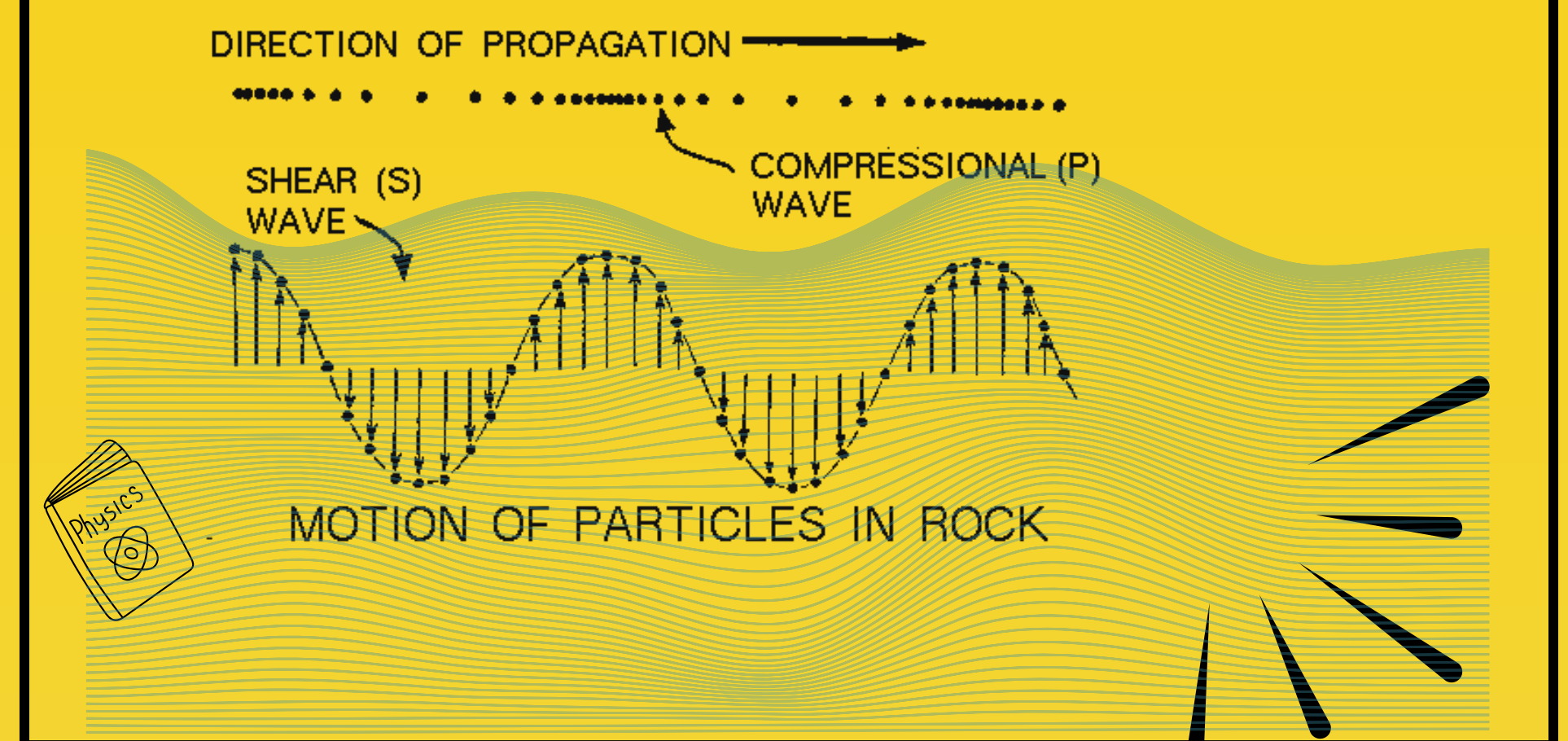
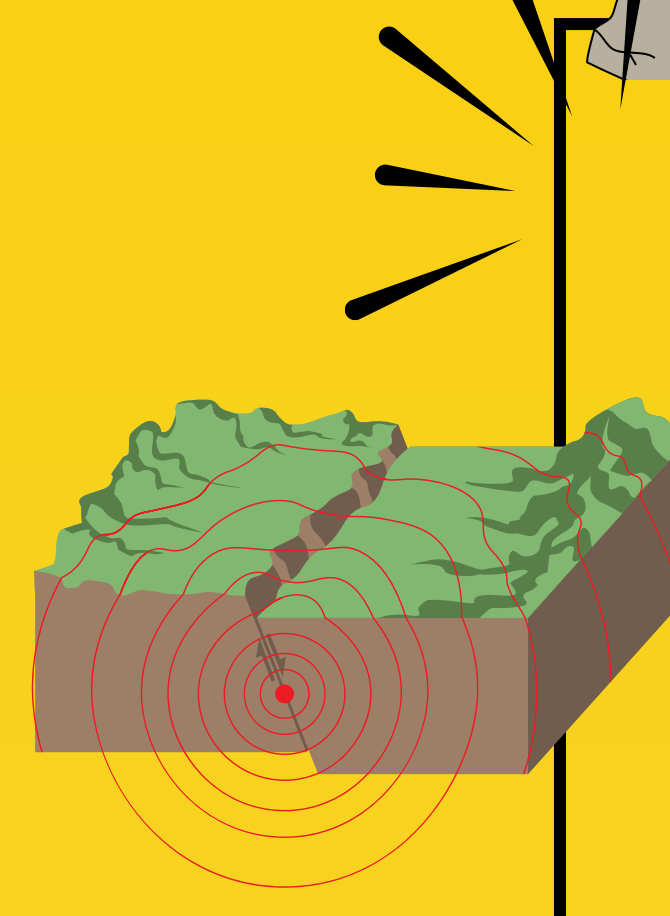
The Shinbashira is heavier than the Tower, therefore behaving differently when shaken by an earthquake, allowing them to cancel each others vibrations out.



Physics behind earthquakes

Earthquakes stem from **tectonic plate** movement where built-up stress along **faults** is suddenly released as **seismic waves**, (waves of energy) causing the ground to shake.

This sudden slip converts stored **elastic energy** into **P-waves** (compression) and **S-waves** (shear), which travel through the Earth and shake the surface as they propagate outward, much like ripples in water.



How did architects manage to overcome these problems?



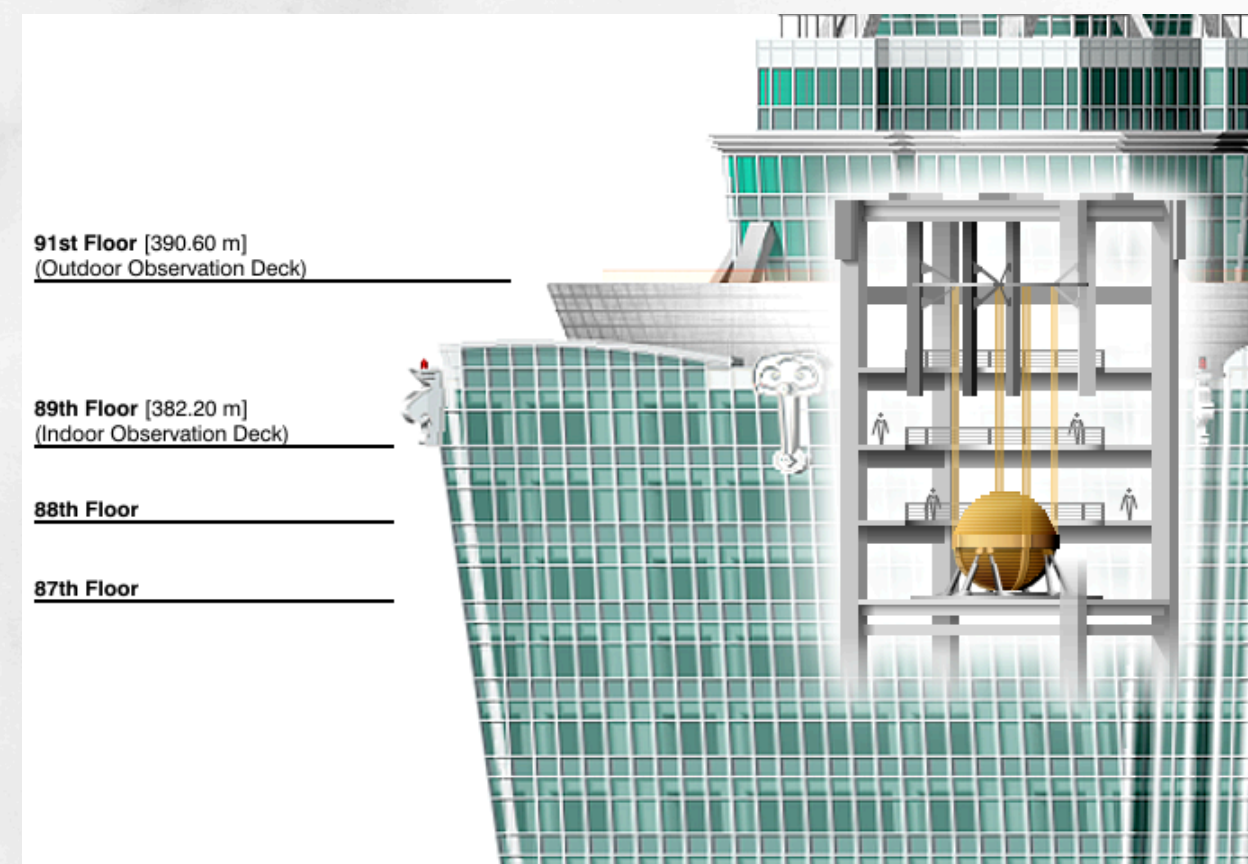
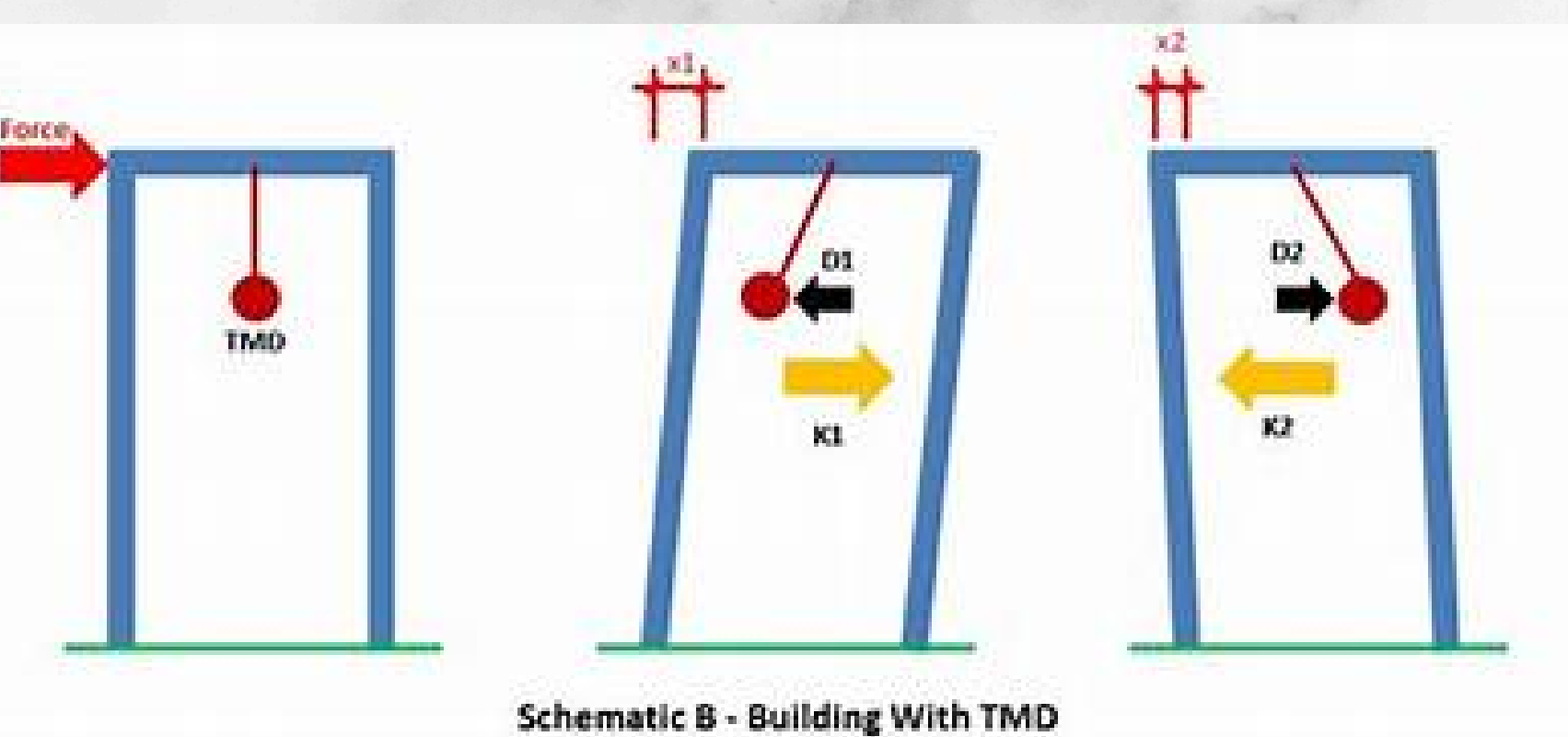
Engineering Methods

Focusing on flexibility, not strength

Engineers designed buildings to absorb seismic energy, rather than resist it

Tuned Mass Dampers (TMDs)

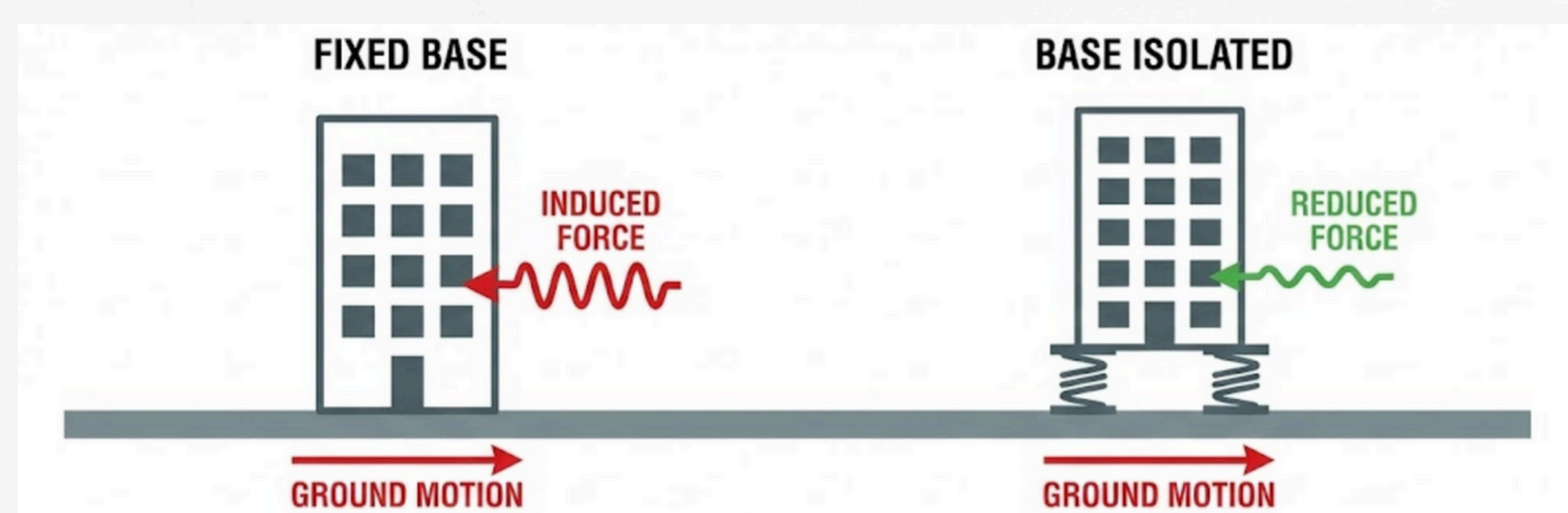
These are devices installed in skyscrapers to counteract swaying caused by seismic waves. A TMD typically consists of a large mass mounted on springs or dampers that move in opposition to the building's motion, reducing oscillations. They work like shock absorbers, using inertia to counteract shaking and reducing structural stress,



Base Isolation

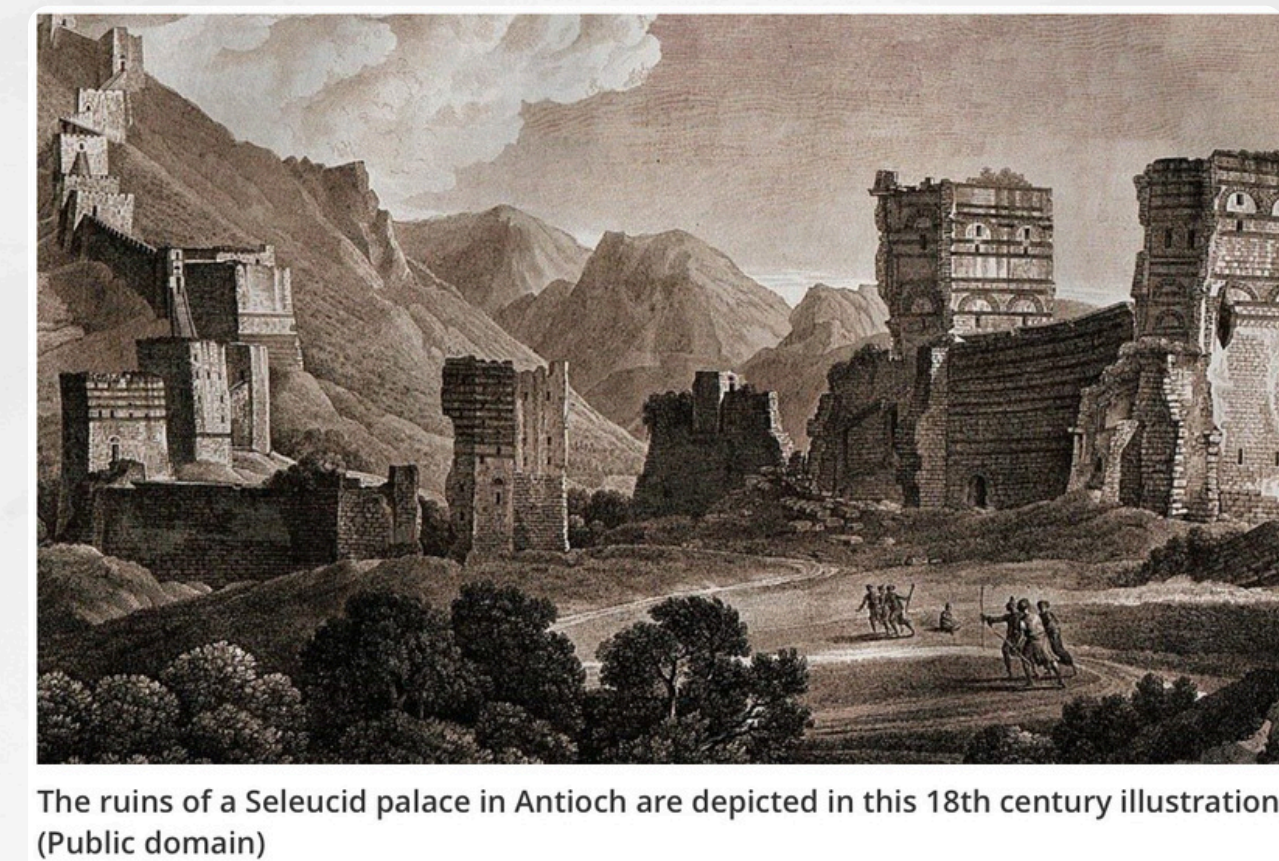
By introducing isolators with low lateral stiffness, the total stiffness of the system decreases significantly. Isolators, flexible bearings in the foundation, allow the building to move independent of ground motion. Therefore the period will be significantly increased

$$T = 2\pi \sqrt{\frac{m}{k}}$$



Earthquake Disaster examples:

- 115CE & 526AD : Antioch (current day where turkey Antakya because it is on top of 3 tectonic plates [1]) ev: Byzantine chornicler John Malas wrote that the earthquake cause a lot of damage to bulidings



2004 Indian Ocean earthquake

- earthquake [2] in ancient Iran (856 AD) caused 200,000 deaths (7.9 magnitude) 5th deadliest earthquake
- 1931 China Earthquakes (Fujian): A series of quakes, including a massive magnitude 7.8, causing widespread collapse and flooding, with death tolls estimated over 200,000.
- 1964 Great Alaska Earthquake (USA): Magnitude 9.2, the second strongest on record, causing tsunamis and extensive damage in Alaska.
- 2004 Indian Ocean Earthquake & tsunami (magnitude 9.1-9.2) killing over 242,000 => poorly constructed buildings collapsing in the middle of the night



References:

<https://www.physics.com.sg/how-skyscrapers-react-to-earthquakes-physics-made-simple>